# CHBB

## FIRST PRINCIPLES FRIDAY

### CLIMATE RISK & CAPITAL

*Why does capital arrive after the damage?*

Start with incentives. Price the risk. Pre-position the response.

# FIRST PRINCIPLE 1 — TIMING IS ECONOMICS

## THE CURRENT SEQUENCE

⚡ **Shock occurs**
Flood · Drought · Hurricane

↓

📋 **Response mobilized**
Weeks of delay

↓

💸 **Funds disbursed**
Months after the shock

↓

📉 **Damage already realized**
Lost output may be irreversible

## THE MECHANISM FAILS

- ✕ Private capital withdraws
- ✕ Sovereign borrowing explodes
- ✕ No pre-positioned liquidity

FIRST-PRINCIPLES QUESTION

**Why wait to finance a predictable risk?**

*The shock is immediate. The financing should be designed before it.*

# FIRST PRINCIPLE 2 — RISK IS NOT THE SAME AS CAPITAL

**0–5%**

Insurance penetration
during extreme events

**0**

Capital markets access
at moment of shock

**Max**

Sovereign funding need
vs. minimum market access

At the moment funding is most needed, access to capital can be weakest.

## LOOK BEYOND THE CAPITAL GAP

The issue may not be a lack of money.
It may be a lack of investable risk structures.

## THE PRICING PROBLEM

Rare events can create catastrophic losses.
Without credible pricing and risk-sharing, markets stay out.



CATALYTIC CAPITAL

## Supports the premium

Can make the risk affordable

→



PRIVATE CAPITAL

## Supplies investable capital

Takes defined, priced risk

→



PLATFORM & GUARANTEE

## Structure and credit support

MDBs can help standardize and de-risk

PRE-AGREED TRIGGER

- ✓ Observable data
- ✓ No case-by-case claims
- ✓ Basis risk explicitly managed

**Liquidity is arranged before the shock.**

# FIRST PRINCIPLE 4 — USE PUBLIC MONEY TO UNLOCK, NOT REPLACE

THE LEVERAGE HYPOTHESIS

**1€**

public / donor capital

↓

**1–4€**

potential private capital mobilized

*The test: does each public euro mobilize additional risk-bearing capital?*

## INVESTORS

→ Earn a contracted return

→ Take a defined layer of tail risk

→ Assess risk, return and impact

## RECIPIENT COUNTRIES

→ Access liquidity when the trigger fires

→ Reduce reliance on emergency borrowing

→ Build a record of resilience

## DONORS

→ Seek more impact per euro

→ Require traceable deployment

→ Fund a repeatable mechanism

# FIRST PRINCIPLE 5 — ALLOCATE LOSSES BEFORE THEY HAPPEN

## WHO BEARS WHICH LOSS?

### SENIOR LAYER

**Private investors**

Takes the contractually defined risk
Receives the agreed return

*→ Protected by junior / first-loss capital*

### JUNIOR / FIRST-LOSS LAYER

**Donor-backed capital**

Absorbs initial losses within agreed limits

## WHY THE STRUCTURE MATTERS

- Risk allocation is explicit
- Investor exposure is contractually defined
- Public capital can improve investability
- Terms must withstand stress scenarios

## THE INVESTMENT TEST

Can the expected return justify
clearly defined loss exposure?

*Validate with independent catastrophe modelling.*

# FIRST PRINCIPLE 6 — A RULE IS ONLY AS GOOD AS ITS DATA

## POSSIBLE PHYSICAL INDICATORS

### Satellite Rainfall

Observed rainfall anomaly

### Soil Moisture Index

Soil moisture and crop stress

### Temperature Anomalies

Temperature deviation from baseline

## POTENTIAL DATA SOURCES

- Copernicus Programme (ESA)
- NOAA — Oceanic & Atmospheric Admin.
- National Meteorological Services

## WHAT A RULE-BASED TRIGGER CAN REDUCE

- Less discretionary decision-making
- Less claims negotiation
- Fewer ad hoc overrides
- Faster payout decisions

**Trigger fires. Contract executes. Capital moves. Basis risk still needs independent review.**

# FIRST PRINCIPLE 7 — EVERY PARTY NEEDS A REASON TO PARTICIPATE

## DONORS

*Danida · European Union · USAID*

→ More impact per euro deployed

→ Auditable use of funds

→ Outcomes that can be measured

## DEVELOPING COUNTRIES

*Global South sovereign issuers*

→ Liquidity when risk materializes

→ Less dependence on emergency debt

→ Build a credible issuance history

## PRIVATE INVESTORS

*Institutional · DFI · ESG mandates*

→ A priced source of return

→ Potential portfolio diversification

→ Impact tied to real risk transfer

## MDBs

*World Bank · African Development Bank*

→ Repeatable market infrastructure

→ A way to mobilize private capital

→ Potential to replicate across regions

Alignment is not automatic. It must be designed, priced and governed.

# FIRST PRINCIPLE 8 — PAYOUT IS NOT THE SAME AS IMPACT

## WHAT THE DESIGN AIMS TO REDUCE

- Delays after a verified shock
- Ad hoc allocation decisions
- Unnecessary process friction
- Uncontrolled transfers
- Weak traceability

→

## A CONTROLLED THREE-LAYER FLOW

**1 — DIRECT SUPPLIER PAYMENT**
Payments route to approved suppliers
Water · Food · Infrastructure
Subject to legal and operational design

**2 — RING-FENCED ACCOUNT**
Funds held separately
Released against documented conditions

**3 — PRE-AGREED USES**
Irrigation · Flood protection
Emergency food supply
Within agreed eligibility rules

The objective is not only to release capital. It is to make the response timely, traceable and useful.

# FIRST PRINCIPLE 9 — TEST THE MODEL BEFORE SCALING

## 01

SOVEREIGN DEMAND

**Target: ≥ 3 letters of intent**

Evidence of issuer interest
and willingness to participate

## 02

CATALYTIC FUNDING

**Target: ≥ 2 commitments**

Coupon / risk support covered
for a defined multi-year horizon

## 03

LEGAL & OPERATIONAL VALIDATION

**Payout mechanics are legally valid**

Operational controls tested
Jurisdiction-specific approval

### GO — all three tests pass

Structure a pilot issuance
Engage anchor investors
Execute a defined mandate

### NO-GO — any critical test fails

Do not deploy capital
Preserve public funds
Reassess once evidence changes

# FIRST PRINCIPLE 10 — TRUST IS PART OF THE PRODUCT



01 GOVERNANCE RISK

## Funds are not used as intended

MITIGATION

Ring-fenced payout design
Supplier controls and audit trail
Independent verification

02 MORAL HAZARD

## Payouts weaken prevention incentives

MITIGATION

Link terms to credible resilience actions
Reward measurable risk reduction

03 MODEL RISK

## Trigger does not match real-world losses

MITIGATION

Use multiple indicators where justified
Calibrate against observed loss data

04 LEGITIMACY RISK

## "Private investors profit from disaster"

MITIGATION

Transparent returns and loss reporting
Public audit trail
Clear explanation of risk transfer

**Good engineering is not enough. Governance determines whether the market is trusted.**

### AFTER-THE-FACT AID

→ reactive

→ delayed

→ limited scale

## PRE-ARRANGED RISK FINANCE

→ pre-funded

→ rule-based

→ designed to scale

Can we move
from rescue

to readiness?

### WHAT CHANGES

Donor capital

→ can mobilize capital

Private capital

→ can take defined risk

Liquidity

→ can arrive at the trigger

Blended finance · Parametric triggers · Controlled payout

*The mechanism must be tested against real-world delivery.*

**Pilot only if demand, funding and legality are proven**

# THE INSTRUMENT — DEFINE THE RISK



INSTRUMENT

## CRB

A catastrophe-bond structure linked to a defined physical climate risk.
Principal is exposed when a pre-agreed event trigger is breached.
No political declaration or claims negotiation; basis risk still requires careful design and review.

ILLUSTRATIVE TAIL-RISK TRANCHE

## ~2.5–3.5% p.a.

An example trigger calibrated to a modelled event probability.
Possible perils: drought · flood · cyclone · heat / water stress.
Pricing depends on expected loss, uncertainty, structure and market conditions.

INVESTOR PROPOSITION

## PRICE THE RISK

Return = reference rate + risk premium.
No trigger: investor earns the contracted return.
Trigger: agreed capital is released according to contract.

**DON'T JUST DEBATE THE RISK. DESIGN A WAY TO PRICE IT.**

# FOLLOW THE CASH FLOWS



AT ISSUANCE

Independent modelling estimates expected loss and trigger probability.
Coupon = reference rate + climate / catastrophe risk premium.
Public support may fund premium, first-loss or guarantee layers.

NO TRIGGER

## MARGIN EARNED

Principal remains intact.
Investor receives the contracted return.
The premium compensates for carrying defined risk.

TRIGGER FIRES

## CAPITAL MOVES

The agreed principal-at-risk is released.
Funds flow through predefined, controlled response channels.
The objective: liquidity at the shock, not months later.

**A credible price can signal how the market assesses modelled physical climate risk.**

# FROM POLICY LABEL TO ECONOMIC MECHANISM



### EU / DONORS

Standardize documentation and eligible triggers.
Use climate / development budgets as catalytic capital.
Tender transparent risk tranches across regions.

### MDB / PLATFORM

Provide independent modelling and verification.
Build issuance and credit-enhancement infrastructure.
Pool geographies and perils where diversification is credible.

### PENSION CAPITAL

Access measurable physical-risk exposure.
Assess diversification on evidence, not assumption.
Tie impact claims to actual contingent capital.

### RECIPIENT

Pre-arranged protection.
Potentially less emergency borrowing.
Rule-based liquidity against verified shocks.
Better terms only where resilience is evidenced.

*The test is the cash-flow mechanism — not the green label.*

# FIRST PRINCIPLES FRIDAY — THE ECONOMIC TEST



CURRENT DEFAULT

## AFTER THE SHOCK

Emergency aid after losses occur.
Allocation can be slow and discretionary.
Borrowing rises when market access is weakest.
Risk may remain unpriced before the event.

ALTERNATIVE TO TEST

## BEFORE THE SHOCK

Capital is arranged before the event.
Pre-agreed triggers can accelerate release.
Public support may mobilize institutional capital.
Defined catastrophe tranches may make some tail risk investable.

ECONOMIC TEST

## THREE TESTS

1. Is the risk independently modelled?
2. Does the price cover expected loss and uncertainty?
3. Does the payout reach the intended response quickly and transparently?

**FIRST PRINCIPLES: PRICE THE RISK. PRE-ARRANGE THE CAPITAL. VERIFY THE OUTCOME.**